

NOETICA MASTER BLUEPRINT

PART 08

V1 Architecture (Scientific Intelligence MVP)

Purpose

The purpose of V1 is not to build the final Noetica platform.

The purpose of V1 is to validate the core hypothesis:

Can Noetica identify and deliver scientifically significant discoveries better than traditional paper-based systems?

V1 focuses on:

- Discovery identification
- Discovery ranking
- Personalization
- Scientific intelligence reports

V1 does not attempt to build:

- Full knowledge infrastructure
- Large-scale knowledge graphs
- AI scientists
- Civilization forecasting

Those belong to future versions.

V1 Success Criteria

A successful V1 should answer:

- Can discoveries be ranked instead of papers?
- Do users find discovery-centric reports valuable?
- Can significance scoring outperform popularity-based ranking?
- Can users discover important developments outside their primary fields?

If these questions are answered positively, Noetica proceeds to V2.

V1 Infrastructure Stack

Automation:

GitHub Actions

Email Delivery:

Gmail SMTP

(App Password Authentication)

Subscriber Collection:

Google Forms

Subscriber Storage:

Google Sheets

Database:

SQLite

Code Hosting:

GitHub

Cost:

₹0/month

Why This Stack Was Selected

GitHub Actions

Advantages:

- Free

- Cloud-hosted
- No VPS required
- No local machine dependency
- Open-source friendly
- Easy scheduling
- Easy contributor onboarding

Purpose:

Daily automation.

Gmail SMTP

Advantages:

- Free
- Easy setup
- Trusted infrastructure
- Suitable for MVP scale

Purpose:

Report delivery.

Google Forms

Advantages:

- No website required
- No backend required
- Fast deployment
- Zero cost

Purpose:

Subscriber acquisition.

Google Sheets

Advantages:

- Simple
- Accessible
- No database management

Purpose:

Store subscriber information.

SQLite

Advantages:

- Lightweight
- Reliable
- Zero maintenance

Purpose:

Store internal intelligence data.

Subscriber Workflow

User

↓

Google Form

↓

Google Sheets

↓

GitHub Actions

↓

Discovery Pipeline

↓

Personalized Report

↓

Gmail SMTP

↓

User Inbox

Subscriber Data Model

Fields:

Email

Name

Expertise Level

Reading Time

Report Frequency

Knowledge Interests

Discovery Preferences

Exploration Preference

Consent Status

Expertise Levels

Beginner

Intermediate

Advanced

Researcher

Industry Professional

Reading Time Options

5 Minutes

10 Minutes

15 Minutes

30 Minutes

60 Minutes

120 Minutes

Report Frequencies

Daily

Weekly

Monthly

Discovery Preferences

Users may select:

Foundational Discoveries

Modern Breakthroughs

Emerging Frontiers

Research Papers

Patents

Research Grants

Startup Funding

Scientific Conferences

Datasets

Open Source Projects

Technology Forecasts

Cross-Disciplinary Insights

Personalization Framework

Inputs:

Interests

Expertise

Reading Time

Frequency

Feedback

Output:

Personalized Scientific Intelligence Report

Forced Exploration Engine

Purpose:

Prevent intellectual echo chambers.

Distribution:

80%

User-selected interests

20%

Outside disciplines

Example

Bioinformatics User

Receives:

Bioinformatics

AI

Genomics

Plus:

Topology

Economics

Philosophy of Science

Materials Science

Potential future connections become visible.

Daily Automation Pipeline

Step 1

GitHub Action Trigger

Schedule:

Daily

Step 2

Collect New Information

Sources:

arXiv

PubMed

OpenAlex

CrossRef

Semantic Scholar

bioRxiv

medRxiv

Step 3

Discovery Extraction

Convert:

Papers



Discovery Candidates

Step 4

Discovery Clustering

Group related evidence.

Example:

CRISPR Cluster

Contains:

Papers

Patents

Datasets

Grants

Products

Step 5

Duplicate Detection

Remove duplicates across sources.

Purpose:

Prevent repeated reporting.

Step 6

Discovery History Update

Track:

First Seen Date

Last Seen Date

Trend Score

Significance Score

Lifecycle Stage

Step 7

Scientific Significance Scoring

Apply Part 04 Framework.

Components:

Evidence

Novelty

Replication

Cross-Field Influence

Knowledge Connectivity

Citation Growth

Funding Momentum

Practical Impact

Step 8

Source Reliability Assessment

Evaluate confidence.

Source Tiers:

Tier 1

Major journals

Major organizations

Tier 2

Peer-reviewed journals

Tier 3

Preprints

Tier 4

Blogs

Media

Tier 5

Social media

Step 9

Emerging Field Detection

Questions:

What fields are appearing?

What fields are merging?

What fields are accelerating?

Step 10

Discovery Lifecycle Classification

Stages:

Speculative

Emerging

Growing

Breakthrough

Established

Foundational

Civilizational

Historical

Step 11

Diversity Enforcement

Prevent domination by one field.

Example:

Not:

18 AI discoveries

2 Biology discoveries

Prefer:

Physics

Biology

Medicine

Mathematics

Economics

AI

Engineering

Philosophy

Step 12

Personalization

Generate user-specific rankings.

Step 13

Report Generation

Create:

Daily Report

Weekly Summary

Monthly Summary

Step 14

Email Delivery

Send via Gmail SMTP.

Step 15

Feedback Collection

Question:

Was this useful?

Options:

Very Useful

Useful

Neutral

Not Useful

Step 16

Feedback Storage

Update personalization profile.

SQLite Database Architecture

Tables:

Subscribers

Discoveries

DiscoveryClusters

DiscoveryHistory

KnowledgeNodes

KnowledgeEdges

SourceReliability

UserFeedback

Reports

EmergingFields

Knowledge Graph Storage (V1)

V1 does not use Neo4j.

Store relationships in SQLite.

Knowledge Nodes

Fields:

Node ID

Node Name

Node Type

Field

Subfield

Knowledge Edges

Fields:

Source Node

Target Node

Relationship Type

Weight

Discovery Cluster Model

Primary Entity:

Discovery Cluster

Not:

Paper

Example

CRISPR

Cluster Contains:

Research Papers

Patents

Clinical Trials

Datasets

Products

Startups

Funding

Discovery Lifecycle Engine

Stages:

Speculative

Emerging

Growing

Breakthrough

Established

Foundational

Civilizational

Historical

Discovery Momentum Engine

Measures:

Research Growth

Funding Growth

Patent Growth

Open Source Growth

Community Growth

Dataset Growth

Ranking Explanation Engine

Every ranking must be explainable.

Example:

Discovery:

Protein Language Models

Score:

91.4

Explanation:

Evidence: 9.2

Novelty: 8.9

Connectivity: 9.5

Funding: 8.4

Cross-Field Influence: 9.7

User Feedback Engine

Purpose:

Continuously improve personalization.

Signals:

Email Opens

Clicks

Ratings

Field Preferences

Ignored Topics

Weekly Intelligence Summary

Contains:

Top Discoveries

Top Trends

Emerging Fields

Most Important Developments

Monthly Frontier Report

Contains:

Most Important Discoveries

Fastest Growing Fields

Cross-Disciplinary Breakthroughs

Technology Convergence Signals

Annual State of Human Knowledge

Contains:

Most Important Discoveries

Most Important Emerging Fields

Most Influential Technologies

Most Influential Datasets

Most Influential Scientific Movements

Forecasts For Next Year

V1 Limitations

Not Included:

User Accounts

Public Dashboard

Patent Intelligence

Grant Intelligence

Organization Intelligence

Researcher Intelligence

Forecasting Engine

API Access

Knowledge Graph Visualization

AI Scientist Layer

Civilizational Impact Engine

These become V2 and V3 responsibilities.

V1 Deliverable

At the end of V1, Noetica should function as:

An automated, evidence-driven scientific intelligence system capable of identifying, ranking, explaining, and delivering the most significant discoveries across the global knowledge ecosystem while remaining completely free to operate.